

Simulation of Phenological Development of Wheat Crops

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ABSTRACT

A wheat phenology model based on the effects of temperature, vernalization and photoperiod was developed by introducing a temperature response function for development rate and vernalization, and the concepts of physiological development days and physiological vernalization days. This model also includes the prediction of the development of leaves, internodes and tillers. It gives EC stage values, i.e. the decimal code defined by Zadoks, J. C., Chang T. T. and Konzak, C. F. (1974) Weed Research 14, 415–421, as outputs. The model was parameterized and tested using observation data from The Netherlands, north and south Germany. Simulation results show that the phenology model can be used to simulate development stages for different wheat varieties with acceptable accuracy. The model can be extended to other cereal crops. © 1998 Elsevier Science Ltd. All rights reserved

INTRODUCTION

Development and growth are two concepts which are often used in crop modelling. Development relates to the physiological age of the plant and to its morphological appearance (Penning de Vries and van Laar, 1982). Growth refers to the increase in weight, volume, length, or area of some part or all of the plant (Ritchie and NeSmith, 1991). The total amount of growth for a given time interval depends largely on photosynthesis, whereas

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partitioning of assimilates to different organs is controlled by plant development stage. Therefore, development simulation is an essential part of a crop model.

For cereal crops, the decimal code system defined by Zadoks *et al.* (1974) has been widely used to record the development stages. The Zadoks stage is a non-linear scale based on irregularly spaced phenological events from sowing to maturity. Using a two-digit code, the Zadoks system can record detailed information about the development status of the plant. However, in many crop models phenological status is reported by identifying the major developmental landmarks (e.g. emergence, jointing), rather than calculating a numerical scale value. This often results in difficulties in comparing the simulated development stages with experimental data. It is desirable to improve crop development models with more detailed and more descriptive stages.

The phenology model developed in this paper takes the above considerations into account. Stages are identified according to the organ development and biomass partitioning of the plant. The model has been finished and parameterized for wheat. It gives the Zadoks stage values, also called EC stages, as model outputs. Because it is developed based on the principle processes (effect of temperature, vernalization and photoperiod) that control the crop development, it can be extended to simulate development of other cereal crops with a few modifications.

MODEL DESCRIPTION

Temperature and photoperiod are the two main factors affecting crop development rate. Water stress may delay or stimulate crop development, but a moderate level of water stress appears to have no direct effect on development (Penning de Vries *et al.*, 1989). In our model, only the effect of temperature and photoperiod on development rate is considered.

The internal stage value that controls partitioning of assimilates is called a VR stage (vegetative/generative stage) value, S_{dev} , which is similar to that used by Penning de Vries *et al.* (1989). For determinate crops it has the value of 0.0 at emergence, 1.0 at anthesis and 2.0 at maturity. Before emergence, a value of -1.0 is set at sowing and -0.5 at germination.

Germination and emergence

Before emergence, the plant grows under the soil surface. The development processes include seed germination and sprout emergence. Germination is simulated using the approach of Jones *et al.* (1986). At germination, S_{dev} is set to a value of -0.5. After germination, the development rate ($R_{\text{dev,emerg}}$) is

controlled by temperature. It is simulated using the linear day-degree system with daily steps of daily mean air temperature (Angus *et al.*, 1981a):

$$R_{\text{dev,emerg}} = (T_{\text{ave}} - T_{\text{base}})0.5 / \sum T \quad [\text{day}^{-1}] \quad (1)$$

where $\sum T$ is the temperature sum needed for emergence ($^{\circ}\text{C day}$). T_{ave} and T_{base} are daily average air temperature and base temperature ($^{\circ}\text{C}$), respectively. Definitions of all the variables and symbols used are given in Appendix A. $\sum T$ is dependent on the sowing depth. In the current model, a constant value of $\sum T$ is used (125 degree-days) according to Groot (1987). Normally this value does not need to be changed, because the sowing depth for one crop does not change very much. If soil temperature is available, it may be used for T_{ave} to improve the model.

Development after emergence

After emergence, a certain number of days, depending on temperature and photoperiod, will be needed for a given genotype to complete its development processes. Under an optimal temperature–photoperiod combination, this number of days is minimized. This minimum number of days for development is defined as total physiological development days (D_s). A unit number of D_s is called one physiological development day. Assuming that the physiological development days needed for vegetative (emergence to anthesis) and reproductive phase (anthesis to maturity) are D_v and D_r ($D_s = D_v + D_r$), the maximum development rate in the two phases $R_{\text{dev,v,max}}$ and $R_{\text{dev,r,max}}$ are the reciprocals of D_v and D_r , respectively. The actual development rate in the vegetative phase ($R_{\text{dev,v}}$) is affected by three variables: temperature, vernalization (of winter-form plants) and photoperiod. In the reproductive phase the development rate ($R_{\text{dev,r}}$) is assumed to be only determined by temperature (for determinate species). A multiplicative relationship is assumed among these factors.

$$R_{\text{dev,v}} = R_{\text{dev,v,max}} f_v(T) f(V) f(h_{\text{php}}) \quad [\text{day}^{-1}] \quad (2)$$

$$R_{\text{dev,r}} = R_{\text{dev,r,max}} f_r(T) \quad [\text{day}^{-1}] \quad (3)$$

where $f_v(T)$, $f_r(T)$ are the temperature response functions of the development rate in the two phases. $f(V)$ and $f(h_{\text{php}})$ account for the effect of vernalization and photoperiod, respectively. Taking the development stage value (S_{dev}) as -1.0 at sowing and -0.5 at germination, the value of S_{dev} n days after germination is expressed as:

$$S_{\text{dev}} = \sum_{d=d_{\text{germ}}}^n R_{\text{dev}} - 0.5 \quad [-] \quad (4)$$

where d_{germ} is the day on which seed germination occurs and the development rate R_{dev} is given as:

$$R_{\text{dev}} = \begin{cases} R_{\text{dev,emerg}} & \text{if } -0.5 \leq S_{\text{dev}} < 0.0 \\ R_{\text{dev,v}} & \text{if } 0.0 \leq S_{\text{dev}} < 1.0 \\ R_{\text{dev,r}} & \text{if } 1.0 \leq S_{\text{dev}} < 2.0 \end{cases} \quad [\text{day}^{-1}] \quad (5)$$

Effect of temperature

The effect of temperature on development rate can be described by using various temperature response curves (Angus *et al.*, 1981b; Penning de Vries *et al.*, 1989; Thornley and Johnson, 1990; Ritchie and NeSmith, 1991; Hodges, 1991; Horie, 1994). It is difficult to give a general equation for various crops. In case no measurement data are available, the temperature response is determined by three cardinal temperatures (minimum, optimum and maximum temperature) in each phase ($T_{\text{min,v}}$, $T_{\text{opt,v}}$, $T_{\text{max,v}}$ for the vegetative phase and $T_{\text{min,r}}$, $T_{\text{opt,r}}$, $T_{\text{max,r}}$ for the reproductive phase) using an introduced temperature response function:

$$= \begin{cases} [f_T(T, T_{\text{min}}, T_{\text{opt}}, T_{\text{max}})] \\ \frac{2(T-T_{\text{min}})^\alpha (T_{\text{opt}}-T_{\text{min}})^\alpha - (T-T_{\text{min}})^{2\alpha}}{(T_{\text{opt}}-T_{\text{min}})^{2\alpha}}; & \text{if } T_{\text{min}} \leq T \leq T_{\text{max}} \\ 0 & ; \text{if } T < T_{\text{min}} \text{ or } T > T_{\text{max}} \end{cases} \quad [-] \quad (6)$$

where the temperature minimum T_{min} , optimum T_{opt} and maximum T_{max} can be replaced by the three cardinal temperatures for the simulated process, respectively. Figure 1 shows the form of eqn (6). If the three cardinal temperatures are known, the parameter α can be calculated by the following equation (let $T = T_{\text{max}}$ and $f_T = 0$):

$$\alpha = \ln 2 / \ln[(T_{\text{max}} - T_{\text{min}})/(T_{\text{opt}} - T_{\text{min}})] \quad [-] \quad (7)$$

If experimental measurement data are available, the model can also use an interpolation function to simulate the temperature response similar to that of Goudriaan *et al.* (1992) and Groot (1987). Otherwise, eqn (6) is used. Some values for maximum development rate and the temperature response of the development rate of several crops in vegetative and reproductive phases can be found in Penning de Vries *et al.* (1989).

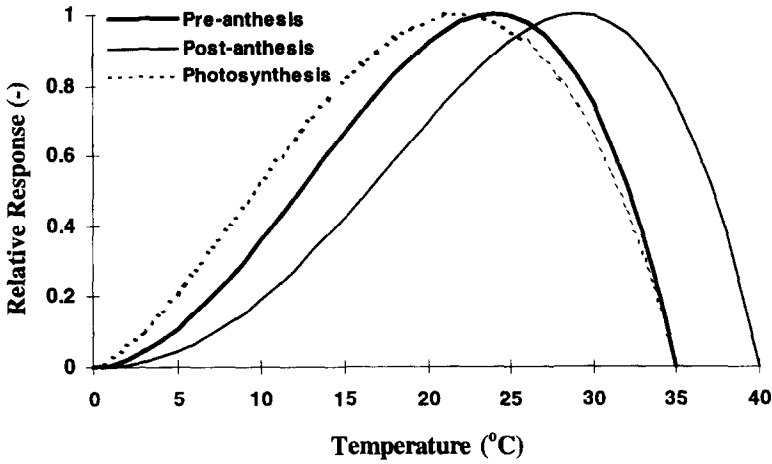


Fig. 1. Temperature response of development rate compared with that of leaf photosynthesis. (Curve parameters: $T_{\min,v} = T_{\min,r} = T_{\min,p} = 0^{\circ}\text{C}$, $T_{\text{opt},p} = 22^{\circ}\text{C}$, $T_{\text{opt},v} = 25^{\circ}\text{C}$, $T_{\text{opt},r} = 30^{\circ}\text{C}$, $T_{\max,p} = T_{\max,v} = 35^{\circ}\text{C}$, $T_{\max,r} = 40^{\circ}\text{C}$.)

Vernalization

Vernalization refers to the low temperature requirement of the plant before its spikelet formation can begin (Ritchie, 1991). It has a low optimum temperature around 0°C . In the model, vernalization is assumed to begin after germination and to finish when the duration of certain low temperatures is long enough (Weir *et al.*, 1984; Ritchie, 1991). The length of the duration is minimized if the temperature is optimal for the vernalization process. This minimum duration, expressed in days, is defined as physiological vernalization days (abbreviated as PVDs), noted as V_{nd} . Assuming that the minimum, optimum and maximum temperature for vernalization is $T_{\min,vn}$, $T_{\text{opt},vn}$ and $T_{\max,vn}$, at a given temperature T , the vernalization rate, $f_{vn}(T)$, can also be simulated using eqn (6). If data are available, this function can also be replaced by the linear interpolation function (reference temperature values can be found in Weir *et al.* (1984); Reinink *et al.* (1986); Ritchie (1991)). The vernalization days accumulated until day n after germination are calculated as:

$$V_n = \sum_{d=d_{\text{germ}}}^n f_{vn}(T) \quad [\text{PVDs}] \quad (8)$$

The vernalization effect on development rate is simulated in a similar way to that by Weir *et al.* (1984). V_n accumulates from germination, while the vernalization effect on development rate begins at emergence. After emergence, development begins only after a minimum value of vernalization days (V_{nb})

is reached. V_{nb} is called base vernalization days. Vernalization is finished after the total vernalization days reaches a value V_{nd} —the vernalization requirement. Both V_{nb} and V_{nd} are genetic dependent. The vernalization function is given:

$$f(V) = \min\{1, \max[0, (V_n - V_{nb})/(V_{nd} - V_{nb})]\} \quad [-] \quad (9)$$

For cereals, vernalization is considered to have ended at floral initiation. When V_{nd} is still not fulfilled at floral initiation, the modifying effect of $f(V)$ is maintained until anthesis. After anthesis, no effect of incomplete vernalization upon the rate of development is assumed (Groot, 1987).

Photoperiodism

Photoperiod affects the rate of development for many crops. Four major groups of plants can be categorized: short-day plants, long-day plants, day-neutral plants, and one dual photoperiod response, the short-long-day plants. It is likely that the temperate cereal grains—wheat, barley, oat, rye—are all long-day plants, and maize, sorghum, rice and soybeans are short-day plants (Major and Kiniry, 1991).

Angus *et al.* (1981*b*) tried several equations to simulate photoperiodic response of spring wheat and found that the negative exponential equation best described the response. Stapper (1984) and Horie (1994) also used such an equation to simulate photoperiodic response of wheat, rice, soybean and barley.

$$f(h_{php}) = 1 - \exp[-\chi\omega'(h_{php} - h_{pc})] \quad [-] \quad (10)$$

where $0 \leq f(h_{php}) \leq 1.0$ $\chi=1$ for long-day plants, $\chi=-1$ for short-day plants. h_{php} is the photoperiod in hours. Because civil twilight is also effective for photoperiodic response, it is included in h_{php} (see Goudriaan and van Laar (1978), van Keulen *et al.* (1982), Stapper (1984)). h_{pc} is the critical photoperiod at which no development occurs. ω' is the photoperiod sensitivity factor. Stapper (1984) derived the photoperiod response for three standard cultivars of wheat from observed development data and found that both ω' and h_{pc} are variety-dependent.

Non-optimal daylengths decrease the development rate. The maximum optimal photoperiod for short-day plants, or minimum optimal photoperiod for long-day plants, is the threshold between optimal and non-optimal daylengths, noted as h_{opp} here. By analysing various published data, Major and Kiniry (1991) found that rice cultivars had h_{opp} varying from less than 10 to more than 13 h. The h_{opp} was consistently 12.5 h for maize and 12 ± 1 h for sorghum of several maturity types. Mungbeans had an h_{opp} that varied from

12 to more than 14 h. The mean h_{opp} value for a number of long-day crops, including wheat, barley, oat, rye, flax, and rape, was 17.7 h and appeared not to vary among species or cultivars. The h_{opp} for 10 oilseed rape cultivars also did not vary and the mean value was 17.1 h (Major and Kiniry, 1991).

Based on the more stable h_{opp} , the photoperiod response function, eqn (10) can be changed to the following equation. It is assumed that the value of $f(h_{php})$ reaches 0.98 when h_{php} reaches h_{opp} .

$$f(h_{php}) = 1 - \exp[-\chi\omega(h_{php} - h_{pc})] \quad [-] \quad (11)$$

$$\omega = 4.0/|(h_{opp} - h_{pc})| \quad [-] \quad (12)$$

$$h_{pc} = h_{opp} - \chi 4.0/\omega \quad [h] \quad (13)$$

For simplicity, h_{opp} is set to constant. The different photoperiod response between varieties can be simulated by only modifying the sensitivity factor ω . In the model, a value of h_{opp} is set to 20 h, the same value as found by Ritchie (1991) and that used in the CERES-Wheat model.

Development of leaves, nodes and tillers

For cereals, the above-ground organs of the plants are leaves, stems, nodes, tillers, ears, flowers and grains. The number of leaves, nodes, tillers on the main stem and the emerging time of ears and flowers are easily recognizable under field conditions. This information can reflect roughly the internal physiological mechanisms and, therefore, can be used to direct crop management. A model for simulating the leaf initiation, leaf emergence, tiller formation of cereals was developed.

Initiation and appearance of leaves

Leaf initiation of cereals begins after emergence and lasts until floral initiation. The rate of leaf appearance is closely related to the development rate toward anthesis. Once final leaf number is determined, these leaves emerge before the flower becomes visible and anthesis occurs (Kiniry *et al.*, 1991). Leaf development is related to the thermal effect of temperature and the effect of photoperiod. Leaf initiation and appearance are assumed to have the same temperature response with an optimum temperature of $T_{opt,lf}$. The maximum and minimum temperatures for leaf initiation and appearance are assumed to be the same as that for the development rate in the vegetative

phase, respectively. It is assumed that photoperiod affects leaf initiation and appearance rate in the same way as for development. Vernalization seems to have no influence on leaf appearance (Cao and Moss, 1991; Mosaad *et al.*, 1995). It is not included in the model.

Assuming the maximum rate of leaf initiation is $R_{\text{prim,max}}$, the actual rate of leaf primordia initiation ($R_{\text{prim,ini}}$) is affected by temperature and photoperiod:

$$R_{\text{prim,ini}} = R_{\text{prim,max}} f_T(T, T_{\text{min,v}}, T_{\text{opt,lf}}, T_{\text{max,v}}) f(h_{\text{php}}) \text{ [primordia day}^{-1}\text{]} \quad (14)$$

Using the leaf collar appearance instead of Haun leaf number (Haun, 1973), the leaf appearance rate ($R_{\text{lf,app}}$) before floral initiation is then given by:

$$R_{\text{lf,app}} = R_{\text{lf,app,max}} f_T(T, T_{\text{min,v}}, T_{\text{opt,lf}}, T_{\text{max,v}}) f(h_{\text{php}}) \text{ [leaves day}^{-1}\text{]} \quad (15)$$

where $R_{\text{lf,app,max}}$ is the maximum rate of leaf appearance. It is smaller than $R_{\text{prim,max}}$. Using the above two rates, the total leaf number (A_{prim}), the emerged and unemerged leaves ($A_{\text{lf,f}}$ and $A_{\text{lf,u}}$) at floral initiation (d_{fi}) can be calculated.

After floral initiation, the rate of leaf appearance is controlled by the development rate ($R_{\text{dev,v}}$):

$$R_{\text{lf,app}} = R_{\text{dev,v}} A_{\text{lf,u}} / S_{\text{fi,flg}} \text{ [leaves day}^{-1}\text{]} \quad (16)$$

where $S_{\text{fi,flg}}$ is the development stage value between floral initiation and flag leaf. At a given day (d), the main stem leaf number (A_{lf}) is then:

$$A_{\text{lf}} = \sum_{d=d_{\text{emerg}}}^d R_{\text{lf,app}} \text{ [leaves]} \quad (17)$$

Internode development

At about terminal spikelet, the first node on the main stem appears and rapid stem elongation begins. After that, the rate of internode appearance is simply related to the development rate in the model. This assumes that the internode appearance process has the same temperature response as the phenological development. With R_{node} and $R_{\text{node,max}}$ representing the actual and maximum internode appearance rate, the relation can be expressed as:

$$R_{\text{node}} = R_{\text{node,max}} f_v(T) f(h_{\text{php}}) \text{ [day}^{-1}\text{]} \quad (18)$$

Tillering

Tillering is simulated in a very simple way, because the number of tillers in the present model is only used for calculating the EC stages and it affects neither the dry weight production, nor grain weight and nitrogen uptake. The potential tiller appearance rate is related to the leaf appearance rate. After three leaves, tillers may be produced in direct proportion to the leaf number. According to Ritchie *et al.* (1987), the main stem tiller number ($A_{\text{til},\text{main}}$) and the total tiller number (A_{til}) can be estimated as:

$$A_{\text{til},\text{main}} = A_{\text{lf}} - 2.5 \quad [\text{tillers stem}^{-1}] \quad (19)$$

$$A_{\text{til}} = A_{\text{til}} + R_{\text{lf},\text{app}}(A_{\text{lf}} - 2.5) \quad [\text{tillers plant}^{-1}] \quad (20)$$

where A_{til} is the potential tiller number a plant can produce. This number will be reduced due to the competition limitations of tillers per m^2 . The total demand of all tillers for assimilate ($C_{\text{dem},\text{til}}$) is estimated as:

$$C_{\text{dem},\text{til}} = 10A_{\text{plt}}A_{\text{til}}\xi_{\text{tiller}}S_{\text{dev}}^3 \quad [\text{kg ha}^{-1}] \quad (21)$$

where A_{plt} is the number of plants per m^2 , ξ_{tiller} is the weight of a single tiller when elongation ceases (g). It is a genetic specific parameter of the crop. S_{dev} is the VR development stage value.

Based on the assimilate supply, the tiller number per m^2 ($A_{\text{til},\text{sq}}$) is then reduced as:

$$A_{\text{til},\text{sq}} = A_{\text{plt}}A_{\text{til}}\min(1, W_{\text{total},\text{st}}/C_{\text{dem},\text{til}}) \quad [\text{tillers m}^{-2}] \quad (22)$$

where $W_{\text{total},\text{st}}$ is the total stem weight (kg ha^{-1}). Tillering is assumed to stop when rapid elongation growth begins (terminal spikelet).

MODEL PARAMETERIZATION AND THE EC STAGE VALUES

Most of the model parameters were determined based on literature data. All the parameter values used in the simulation and the data sources based on which these parameter values are determined are listed in Table 1. In order to test the model using experimental data, three winter wheat varieties from several experiments in different geographical locations were used. The name of the locations, the experimental years and the wheat genotypes used in each location are listed in Table 2. Among these experimental data, the data from The Netherlands (Randwijk, Nagele and Lelystad) have been used to

TABLE 1
The Parameter Values Used in the Development Model

	<i>Symbol</i>	<i>Unit</i>	<i>Value</i>	<i>Determined or estimated based on data sources</i>
Pre-anthesis development	D_v	PDDs	38	Angus <i>et al.</i> , 1981 <i>b</i> ; Groot, 1987
	$T_{\min,v}$	°C	0	van Keulen and Seligman, 1987
	$T_{\text{opt},v}$	°C	24	Angus <i>et al.</i> , 1981 <i>b</i> ; Versteeg and van Keulen, 1986
	$T_{\max,v}$	°C	35	van Keulen and Seligman, 1987; Groot, 1987
Post-anthesis development	D_r	PDDs	25 (<i>cv.</i>)	Angus <i>et al.</i> , 1981 <i>b</i> ; Groot, 1987
	$T_{\min,r}$	°C	0	van Keulen and Seligman, 1987
	$T_{\text{opt},r}$	°C	29	assumed 5°C higher than $T_{\text{opt},v}$
	$T_{\max,r}$	°C	40	assumed 5°C higher than $T_{\max,v}$
Vernalization	V_{nd}	PVDs	46 (<i>cv.</i>)	Groot, 1987; Geisler, 1988
	V_{nb}	PVDs	$0.2V_{nd}$	assumed
	$T_{\min,vn}$	°C	-1	Weir <i>et al.</i> , 1984; Geisler, 1988; Reinink <i>et al.</i> , 1986; Ritchie, 1991
	$T_{\text{opt},vn}$	°C	2	Weir <i>et al.</i> , 1984; Geisler, 1988; Reinink <i>et al.</i> , 1986; Ritchie, 1991
	$T_{\max,vn}$	°C	15	Weir <i>et al.</i> , 1984; Geisler, 1988; Reinink <i>et al.</i> , 1986; Ritchie, 1991
Leaf, nodes and tiller development	$R_{\text{prim,max}}$	primodia day ⁻¹	1.0	assumed
	$R_{\text{lf,app,max}}$	leaves day ⁻¹	0.30	McMaster and Wilhelm, 1995; Slafer and Rawson, 1995; Frank and Bauer, 1995
	$T_{\text{opt,lf}}$	°C	22	Slafer and Rawson, 1995
	$R_{\text{node,max}}$	Node ⁻¹	0.28	Groot, 1987
	ξ_{tiller}	g stem ⁻¹	2.0 (<i>cv.</i>)	Godwin <i>et al.</i> , 1990
VR stage at floral initiation	$S_{\text{dev,fi}}$	—	0.20	van Keulen and Seligman, 1987; Groot, 1987

cv. indicates the values are cultivar-dependent. The given values followed by (*cv.*) are typical values for various cultivars.

estimate D_v and to obtain the relationship between EC stage values and VR stage values.

It is assumed that the minimum and maximum temperatures for development in the pre-anthesis phase (from emergence to anthesis) are equal to those for photosynthesis (Fig. 1) and the optimum temperature for this phase is 2°C higher than that for photosynthesis. This assumption is based on the comparison of data of Angus *et al.* (1981*b*), van Keulen and Seligman (1987), Groot (1987) and Slafer and Rawson (1995). The optimum and maximum temperatures for development in the post-anthesis phase (anthesis to

TABLE 2

The Names and Positions of the Locations Where the Experimental Measurements Are Used to Test the Phenology Model of SPASS-Wheat, the Genotypes Used in the Experiments and the Weather Station Names Where the Weather Data Are Used in the Simulation

	Location/ field name	Latitude (degree)	Longitude (degree)	Altitude (m)	Year	Genotype name	Weather station name
North Germany	Intensive Loam Site	52	10	145	1988–91	Kanzler	Neuenkirchen
	Intensive Loam Site	52	10	145	1990–91	Orestis	Neuenkirchen
	Bockschlag	52	10	145	1991	Kanzler	Neuenkirchen
South Germany	Scheyern	48	11	497	1990–91	Orestis	DWD Huell
The Netherlands	Randwijk	51.95	5.75	–	1982–84	Aminda	Wageningen
	Nagele	52.62	5.75	–	1982–84	Aminda	Swifterbant
	Lelystad	52.50	5.50	–	1982–84	Aminda	Swifterbant

maturity) are shifted 5°C higher than those for the pre-anthesis phase to indicate that higher temperatures are needed in the reproductive phase (Fig. 1).

The minimum threshold of the optimal photoperiod (h_{opp}) of wheat is set to 20 h, that is the same value as found by Ritchie (1991) and used in the CERES-Wheat model, 2.3 h longer than the value of Major and Kiniry (1991). The base vernalization days V_{nb} are assumed to be one fifth of the saturated vernalization days.

The maximum leaf appearance rate is set to 0.30 leaves day⁻¹ based on the data of McMaster and Wilhelm (1995), Slafer and Rawson (1995) and Frank and Bauer (1995). The temperature response is also simulated using eqn (6) with three cardinal temperatures listed in Table 1. The calculated appearance rates at different temperatures approximately represent the maximum values of Slafer and Rawson (1995) (Fig. 2).

In the current model, the value of D_v is set to be constant, while the value of D_r is variety-dependent. Using the Dutch experimental data of Groot (1987) for winter wheat cultivar Arminda, together with the temperature responses described previously, the development stage values were simulated for all the sowing dates in the three locations from 1982 to 1984. It was found that 38 and 25 days were the suitable values for D_v and D_r , respectively. In NWHEAT (Groot, 1987), the original base photoperiod (critical photoperiod) was set to be 0 h before double ridge and 7 h after double ridge. Assuming that the cultivar has the V_{nd} and V_{nb} values as suggested by Groot (1987), a value of 0.3 for ω (critical photoperiod 6.6 h) can be chosen for the photoperiod response of cultivar Arminda.

Four parameters are cultivar-dependent (Table 3). There are no measured values for these genetic parameters. Their values were estimated based on the

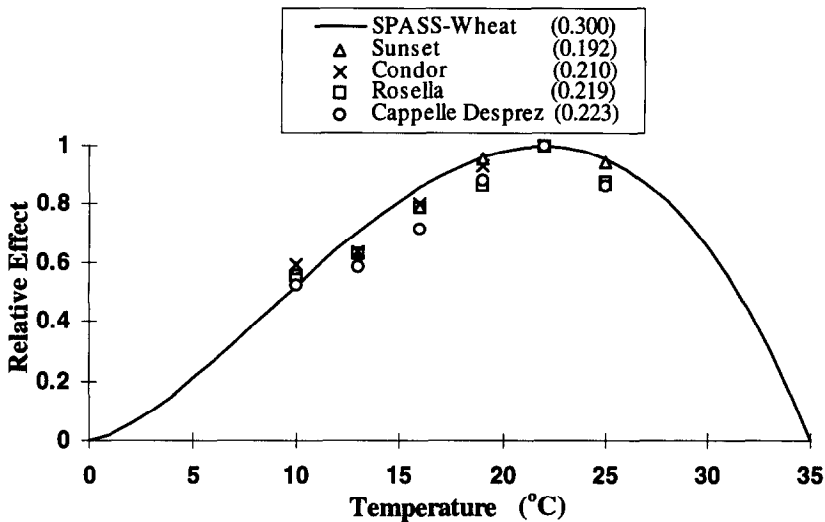


Fig. 2. The temperature response curve for leaf appearance rate in SPASS-Wheat and the measured leaf appearance rate by Slafer and Rawson (1995) for wheat varieties Sunset, Condor, Rosella and Cappelle Desprez. The values in the blanks indicate the maximum leaf appearance rate (leaves day⁻¹). The minimum, optimum and maximum temperatures are 0, 22 and 35°C, respectively.

length of the pre-anthesis and post-anthesis phases and by using the genotype information summarized by the Federal Variety Office of Germany (Bundessortenamt, 1994) for the two German varieties Kanzler and Orestis. The parameter values for the three wheat varieties in the simulation are listed in Table 3.

In practice, the EC stages or Zadoks stages (Zadoks *et al.*, 1974) are mostly used. It is useful to give the EC stage values as model outputs. However, there is no direct relationship between EC and VR stage values. VR stage is a continuous value describing the progress of development, while EC stage is a discontinuous code corresponding to the organ occurrence. In order to translate the VR stage value to EC stage values, simulations were run for all the Dutch data with the parameter values described previously. For each recorded EC stage value of the main stem, a VR stage value was

TABLE 3
Genetic Parameter Values Used for Simulating the Three Wheat Varieties

	ω	D_r	V_{nd}	ξ_{tiller}
Arminda	0.3	23	46	2.0
Kanzler	0.3	25	46	2.0
Orestis	0.25	25	46	2.0

obtained on the same date. Then for the same EC stage value, an average VR stage value was calculated. Based on this calculation and the results of van Keulen and Seligman (1987), Table 4 was established.

The VR stage value at floral initiation was assumed to be 0.20. Floral initiation is defined as the first appearance of the double ridges (van Keulen and Seligman, 1987). According to Groot (1987), floral initiation occurs at a VR stage value of approximately 0.194 (floral initiation and anthesis appeared 145 and 747 degree-days after emergence, respectively). van Keulen and Seligman (1987) found that the ratio of the temperature sum between emergence and floral initiation to that between emergence and anthesis for spring wheat ranges from 0.18 to 0.24.

In the model, after VR stage 0.65 the EC values are directly related to VR values, i.e. the EC value increases continuously with the VR value from one stage to the next (e.g. if VR changes from 0.65 to 0.90, EC changes from 40 to 50). Before VR stage 0.65, the EC stage values are estimated from the VR stage values and the number of main stem leaves, tillers and internodes as follows:

$$\text{if } S_{\text{dev}} < 0.0, S_{\text{EC}} = 10(1 + S_{\text{dev}})$$

$$\text{if } 0.0 \leq S_{\text{dev}} < 0.45 \text{ and } A_{\text{mtil}} = 0, S_{\text{EC}} = 10 + A_{\text{mlv}}$$

$$\text{if } 0.0 \leq S_{\text{dev}} < 0.45 \text{ and } A_{\text{mtil}} > 0, S_{\text{EC}} = 20 + A_{\text{mtil}}$$

$$\text{if } 0.45 \leq S_{\text{dev}} < 0.65, S_{\text{EC}} = 30 + A_{\text{mnd}}$$

TABLE 4
Translation of VR Stage Values to EC Stages

<i>Phase</i>	<i>Start of process</i>	<i>Start of stage</i>	<i>VR stage</i>	<i>EC stage</i>
Pre-emergence	germination	sowing	-1.00	00
	emergence	germination	-0.50	05
Pre-anthesis (vegetative)	leaf initiation	emergence	0.00	10
	spikelet initiation	floral initiation	0.20	14-22
	rapid stem elongation	terminal spikelet	0.45	30
	spike growth	flag leaf	0.65	40
		heading	0.90	50
Post-anthesis (reproductive)	pollination	anthesis	1.00	60
	grain filling	milk development	1.15	70
		dough development	1.50	80
Ripening	ripening	ripening	1.95	90
		maturity	2.0	92

where S_{dev} is the VR stage value, S_{EC} the EC stage value, A_{mlv} the main stem leaf number, A_{mtl} the main stem tiller number and A_{mnd} the main stem node number, respectively.

SIMULATION RESULTS

The phenology model was tested using experimental data described previously. Although the data from The Netherlands (Randwijk, Nagele and Lelystad) are not independent data for the model, comparison of the simulated and observed development stages using these dependent data shows the fitness of the model. The German data are absolutely independent. A comparison of the simulated and observed values using these independent data shows the prediction ability of the model.

Simulation results using The Netherlands data

The simulation results using The Netherlands data are shown in Figs 3 and 4. A detailed description of these experimental data was given by Groot (1987) and Groot and Verberne (1991). Because one EC stage can last several days and most observations were made periodically (weekly, for example), the date on which one EC stage was recorded may not be the first occurring date of this EC stage. If the model gives the same EC stage value as the observed one on a day, then the model is considered to function well. Figure 3 shows that the simulated EC stage values can match the observed values very well. The slope of the regression line is very close to 1.0 (0.968) and the y intercept is very small (1.321). The coefficient of determination (R^2) indicates that more than 97% of the total variance was explained by the model.

Assuming the date on which an EC stage value was recorded is the first date on which the EC stage occurred, the simulated and observed occurring dates of each EC stage values are compared in Fig. 4. It can be seen that the simulated values also match the observed ones very well. The regression line is very close to 1:1 and the y intercept is very small. The coefficient of determination (R^2) indicates that approximately 97% of the total variance can also be explained by the model for the occurring dates of each EC stage values.

In Fig. 4, the values for the early stages are scattered. This means that the early dates of EC stage were not well simulated. However, in Fig. 3, the simulated EC stages matched the observed ones very well in the early stage, except the first observation in each experiment. The reason for the scattering in Fig. 4 may be that the dates on which EC stages were recorded were not the first occurring dates of these stages, because of the rapid development and long interval between observations.

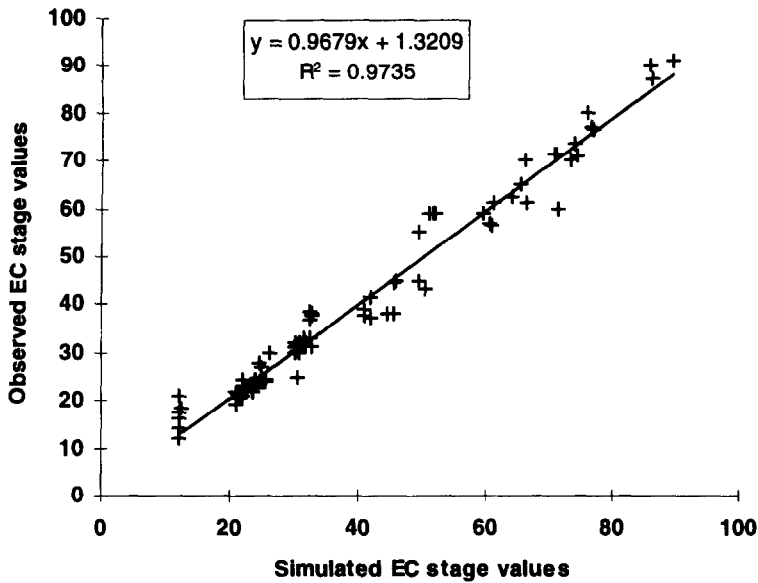


Fig. 3. Comparison of simulated and observed EC stage values for winter wheat variety Aminda in The Netherlands. Data are for 2 years (1982–84) at three locations (Randwijk, Nagele and Lelystad).

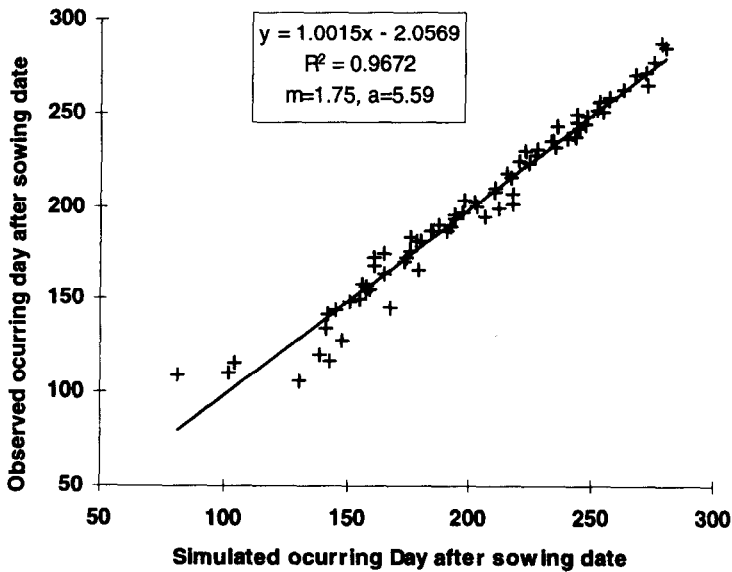


Fig. 4. Comparison of simulated and observed occurring dates for various EC stage values for winter wheat variety Aminda in The Netherlands. Data are for 2 years (1982–84) at three locations (Randwijk, Nagele and Lelystad).

Simulation results—Braunschweig, north Germany, 1988–91

The phenology data from two research catchments in north Germany (original data from McVoy *et al.* (1995)) were also used to test the phenology model. The experiment was conducted in two fields (the Intensive Loam Site and Bockschlag) near Braunschweig with two winter wheat varieties: Kanzler and Orestis. Simulations were run for the Intensive Loam Site for variety Kanzler in growing season 1988–89 and for variety Orestis in growing season 1990–91. At Bockschlag, only the variety Kanzler was grown.

Comparisons of the simulated and measured EC stage values are given in Figs 5–7. The simulated and observed EC stage values were in close agreement. The simulated and observed days after sowing for each observed EC stage of variety Kanzler in Bockschlag are listed in Table 5. For each observed EC stage, a time interval of simulated values is given. During each time interval, the simulated EC values indicate only one EC stage, e.g. EC 22 lasts 8 days from day 138 to day 146 in the simulation (Table 5). The minimum difference in Table 5 means the difference in days between the appearance date of an observed value and the nearest date on which the same simulated value occurs. The appearance difference means the difference in days between the observed date and the first occurring date of the corresponding simulated value. Assuming all observations are on the first appearance dates of various stages, the simulated and observed appearance dates are compared in Fig. 8. It can be seen from both Table 5 and Fig. 8

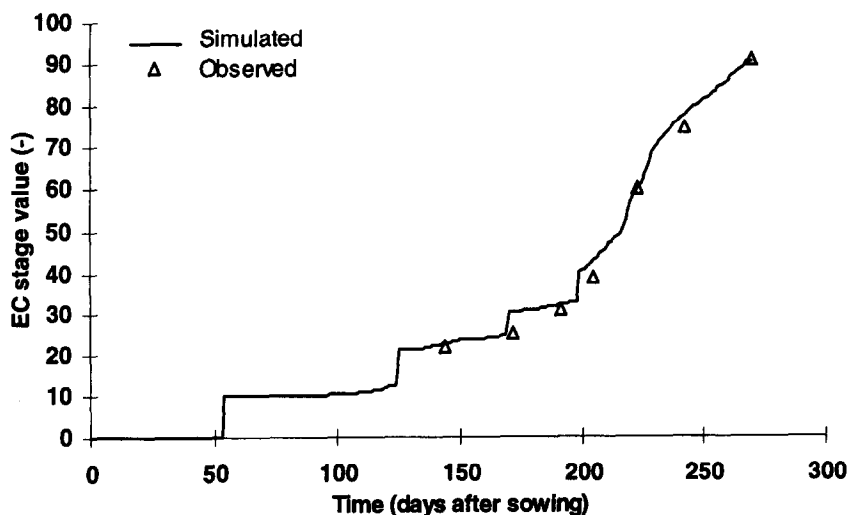


Fig. 5. Comparison of simulated and observed EC stage values of winter wheat variety Kanzler in the growing year 1990–91 at Bockschlag, north Germany.

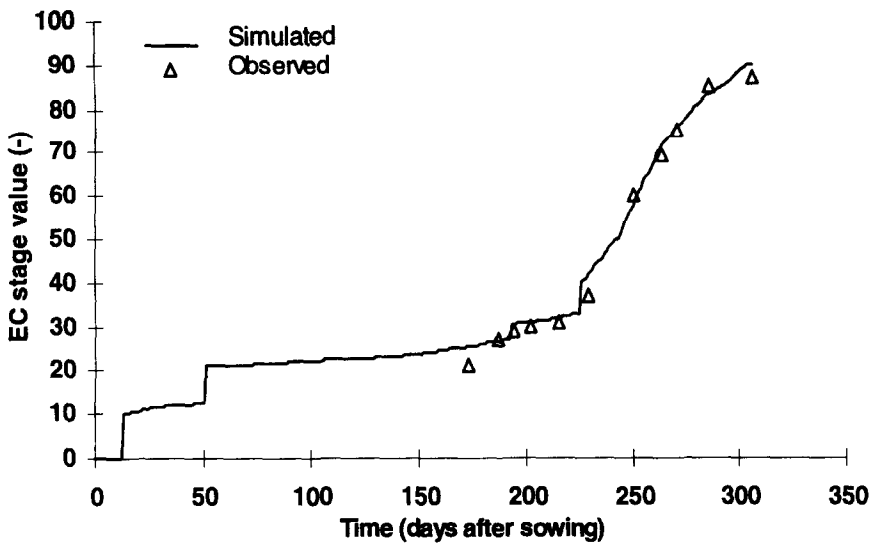


Fig. 6. Comparison of simulated and observed EC stage values of winter wheat variety Kanzler in the growing year 1988-89 at the Intensive Loam Site, north Germany.

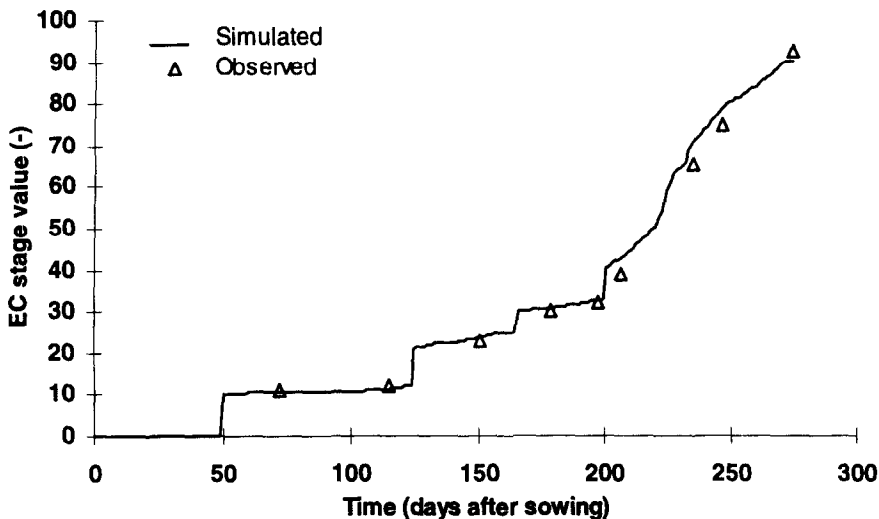


Fig. 7. Comparison of simulated and observed EC stage values of winter wheat variety Orestis in the growing year 1990-91 at the Intensive Loam Site, north Germany.

that the simulated and observed dates are in good agreement. The average absolute prediction error is about 4.3 days. Table 5 shows that in the observation of early EC stages an observation error of about 5 days may occur. This is even larger than the prediction error.

TABLE 5

The Observed and Simulated Dates for Various EC Stage Values for Wheat Variety Kanzler at Bockschlag (1990–91), North Germany

<i>Stage description</i>	<i>EC code</i>	<i>Observed days after sowing</i>	<i>Simulated days after sowing</i>	<i>Minimum difference</i>	<i>Appearance difference</i>
Main shoot and two tillers	22	144–149	138–146	0	6
Main shoot and five tillers	25	172	170 (EC30)	2	2
Pseudo stem erection	30	172	170–181	2	2
First node detectable	31	191	182–190	1	9
Flag leaf ligule visible	39	205	199–204 (EC40–42)	1	6
Flag leaf sheath extending	41 (–42)	205	199–204 (EC40–42)	1	6
Begin of anthesis	60 (–61)	223	223	0	0
Medium milk	75 (–76)	242	237–240	2	5
Caryopsis hard	91	271	270	1	1

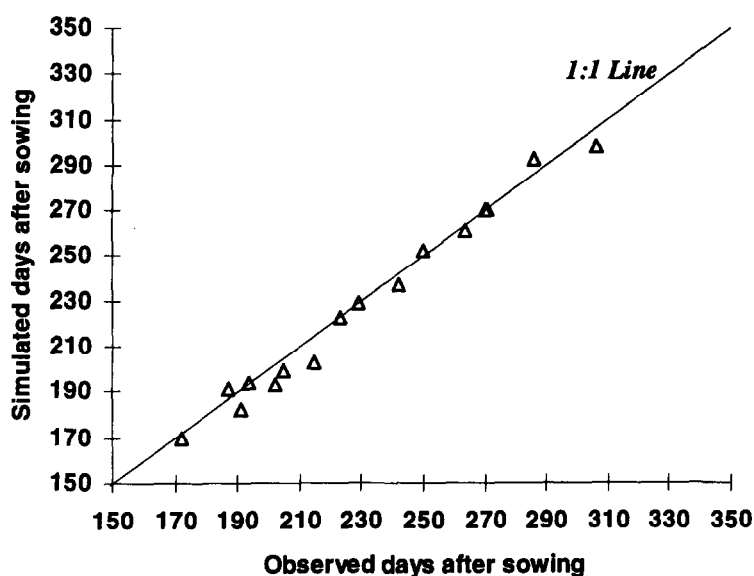


Fig. 8. Comparison of simulated and observed appearance dates of various EC stages of wheat variety Kanzler at Bockschlag (1990–91) and the Intensive Loam Site (1988–89), north Germany.

Simulation results—Scheyern, south Germany, 1990–91

Using experimental data from the Munich Research Network on Agroecosystems (FAM-Forschungsverbund Agrarökosysteme München), the wheat phenology model was tested for the winter wheat variety Orestis in

the growing year 1990–91 in Scheyern near Munich, south Germany. The model simulated the EC stage reasonably well with the same genetic parameters as that used for Orestis in north Germany (Fig. 9). This indicates that the development model is not site-specific and can be used in different geographic regions.

The simulated EC stage 55 occurred 7 days later than the observed stage. This difference may be caused by the calculation method from VR stage to EC stage and by the assumption that EC stage 55 is located at the middle point between VR stage 0.9 and 1.0. Based on the definition of Zadoks *et al.* (1974), EC stage 54 or 55 is reached when half of inflorescence emerged, which is difficult to accurately identify. Observation inaccuracy cannot be excluded either.

DISCUSSION AND CONCLUSION

The phenology model was developed by considering the effects of temperature, vernalization and photoperiod. In the model, a generic temperature response function and the concepts of physiological development day (PDDs) and physiological vernalization day (PVDs) were introduced. The temperature function can be reparameterized by changing the three cardinal temperatures and the model can easily be used for other cereal crops. The physiological development day concept is considered to be easier to understand than the degree-day concept. The physiological development day is the

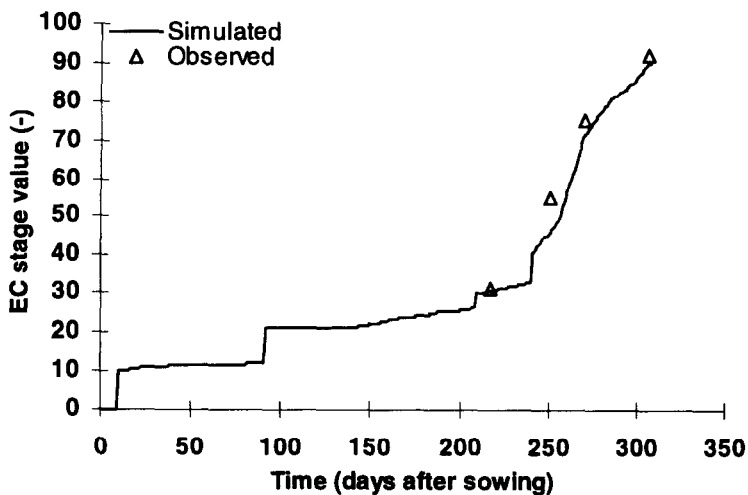


Fig. 9. Comparison of simulated and observed EC stage values of winter wheat variety Orestis in the growing year 1990–91 in Scheyer, south Germany.

time period (in days) during which a plant completes its development processes from emergence to maturity under optimal environmental conditions.

Most existing models have their own predefined internal developmental stage scale values as model outputs. Most internal scales are different from the development stage scales used in practice. This has resulted in problems in comparing the simulated with observed development stages. Our model uses internal stage values (VR stage) within the system, but gives EC stage values as model outputs. This makes simulation results more understandable and simplifies the comparison of simulation results with observed development stages. The model can easily be used for other cereal crops by modifying the parameters.

Simulation results with the wheat data from The Netherlands, north and south Germany indicated that the parameterized developmental model can simulate the EC stages for three wheat varieties with acceptable accuracy. This also indicates that the model can be applied to relatively large geographic regions and is not site-specific.

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REFERENCES

- Angus, J. F., Cunningham, R. B., Moncur, M. W. and Mackenzie, D. H. (1981a) Phasic development in field crops I. Thermal response in the seedling phase. *Field Crops Research* **3**, 365–378.
- Angus, J. F., Mackenzie, D. H., Morton, R. and Schafer, C. A. (1981b) Phasic development in field crops II. Thermal and photoperiodic responses of spring wheat. *Field Crops Research* **4**, 269–283.
- Bundessortenamt (1994) Beschreibendes Sortenliste: Getreide, Mais, Ölfrüchte, Leguminosen, Hackfrüchte. Landbuch-Verlag.
- Cao, W. and Moss, D. N. (1991) Vernalization and phyllochron in winter wheat. *Agronomy Journal* **83**, 173–179.
- Frank, A. B. and Bauer, A. (1995) Phyllochron differences in wheat, barley and forage grasses. *Crop Science* **35**, 19–23.
- Geisler, G. (1988) Pflanzenbau: Ein Lehrbuch-Biologische Grundlagen und Technik der Pflanzenproduktion. 2., neubearbeitete und erweiterte Auflage, pp. 109–111. Verlag paul Parey, Berlin und Hamburg.
- Godwin, D. C., Ritchie, J., Singh, U. and Hunt, L. (1990) A users guide to CERES Wheat-v2.10. Second Edition. Michigan State University, 94 pp.

- Goudriaan, J. and van Laar, H. H. (1978) Calculation of daily totals of the gross CO₂ assimilation of leaf canopies. *Netherlands Journal of Agricultural Science* **26**, 373–382.
- Goudriaan, J., van Keulen, H. and van Laar, H. H. (1992) Crop growth model for potential production (SUCROS1). In *Simulation of Crop Growth for Potential and Water-limited Production Situations (as Applied to Spring Wheat)*, eds H. H. van Laar, J. Goudriaan and H. van Keulen, pp. 27–72. Simulation Reports CABO-TT no. 27, Centre for Agrobiological Research and Department of Theoretical Production Ecology, Wageningen Agricultural University.
- Groot, J. J. R. (1987) Simulation of nitrogen balance in a system of winter wheat and soil. Simulation report CARBO-TT no. 13. Centre for Agrobiological Research (CABO) and Department of Theoretical Production Ecology, Agricultural University Wageningen.
- Groot, J. J. R. and Verberne, E. L. J. (1991) Response of wheat to nitrogen fertilization, a data set to validate simulation models for nitrogen dynamics in crop and soil. *Fertilizer Research* **27**, 349–383.
- Haun, J. R. (1973) Visual quantification of wheat development. *Agronomy Journal* **65**, 116–119.
- Hodges, T. (1991) Temperature and water stress effect on phenology. In *Predicting Crop Phenology*, ed. T. Hodges, pp. 7–13. CRC Press, Boca Raton, Ann Arbor, Boston.
- Horie, T. (1994) Crop ontogeny and development. In *Physiology and Determination of Crop Yield*, eds K. J. Boot, J. M. Bennett, T. R. Sinclair and G. M. Paulsen, pp. 153–180. American Society of Agronomy, Crop Science Society of America, Soil Science Society of America, Madison, Wisconsin.
- Jones, C. A., Ritchie, J. T., Kiniry, J. R. and Godwin, D. C. (1986) Subroutine structure. In *CERES-Maize: a Simulation Model of Maize Growth and Development*, eds C. A. Jones and J. R. Kiniry, pp. 49–111. Texas A and M University Press, Texas.
- Kiniry, J. R., Rosenthal, W. D., Jackson, B. S. and Hoogenboom, G. (1991) Predicting leaf development of crop plants. The CERES-Wheat phenology model. In *Predicting Crop Phenology*, ed. T. Hodges, pp. 29–42. CRC Press, Boca Raton, Ann Arbor, Boston.
- Major, D. J. and Kiniry, J. R. (1991) Predicting daylength effects on phenological processes. In *Predicting Crop Phenology*, ed. T. Hodges, pp. 15–28. CRC Press, Boca Raton, Ann Arbor, Boston.
- McMaster, G. S. and Wilhelm, W. W. (1995) Accuracy of equations predicting the phyllochron of wheat. *Crop Science* **35**, 30–36.
- McVoy, C. W., Kersebaum, K. C., Arning, M., Kleeberg, P., Othmer, H. and Schröder, U. (1995) A data set from North Germany for the validation of agroecosystem models: documentation and evaluation. *Ecological Modelling* **81**, 265–297.
- Mosaad, M. G., Ortiz-Ferrara, G., Mahalakshmi, V. and Fischer, R. A. (1995) Phyllochron response to vernalization and photoperiod in spring wheat. *Crop Science* **35**, 168–171.
- Penning de Vries, F. W. T. and van Laar, H. H. (1982) Simulation of growth processes and the model BACROS. In *Simulation of Plant Growth and Crop Production*, eds F. W. T. Penning de Vries and H. H. van Laar, pp. 114–135. Pudoc, Wageningen.
- Penning de Vries, F. W. T., Jansen, D. M., ten Berge, H. F. M. and Bakema, A. (1989) *Simulation of Ecophysiological Processes of Growth in Several Annual Crops*. Pudoc Wageningen.

- Reinink, K., Jorritsma, I. and Darwinkel, A. (1986) Adaptation of the AFRC wheat phenology model for Dutch conditions. *Netherlands Journal of Agricultural Science* **34**, 1–13.
- Ritchie, J. T. (1991) Wheat phasic development. In *Modeling Plant and Soil Systems*, eds J. Hanks and J. T. Ritchie, pp. 31–54. American Society of Agronomy, Crop Science Society of America, Soil Science Society of America, Madison, Wisconsin.
- Ritchie, J. T. and NeSmith, D. S. (1991) Temperature and crop development. In *Modeling Plant and Soil Systems*, eds J. Hanks and J. T. Ritchie, pp. 5–29. American Society of Agronomy, Crop Science Society of America, Soil Science Society of America, Madison, Wisconsin.
- Ritchie, J. T., Godwin, D. C. and Otter-Nacke, S. (1987) CERES-WHEAT. A simulation model of wheat growth and development. Unpublished model documentation.
- Slafer, G. A. and Rawson, H. M. (1995) Rates and cardinal temperatures for processes of development in wheat: effects of temperature and thermal amplitude. *Australian Journal of Physiology* **22**, 913–926.
- Stapper, M. (1984) SIMTAG: A simulation model of wheat genotypes. University of New England, ICARDA.
- Thornley, J. H. M. and Johnson, I. R. (1990) *Plant and Crop Modelling*. Oxford University Press, Oxford.
- van Keulen, H. and Seligman, N. G. (1987) *Simulation of Water Use, Nitrogen Nutrition and Growth of a Spring Wheat Crop*. Pudoc, Wageningen.
- van Keulen, H., Penning de Vries, F. W. T. and Drees, E. M. (1982) A summary model for crop growth, In *Simulation of Plant Growth and Crop Production*, eds F. W. T. Penning de Vries and H. H. van Laar. Simulation Monographs. Pudoc, Wageningen.
- Versteeg, M. N. and van Keulen, H. (1986) Potential crop production prediction by some simple calculation methods, as compared with computer simulations. *Agricultural Systems* **19**, 249–272.
- Weir, A. H., Bragg, P. L., Porter, J. R. and Rayner, J. H. (1984) A winter wheat crop simulation model without water or nutrient limitations. *Journal of Agricultural Science* **2**, 371–382.
- Zadoks, J. C., Chang, T. T. and Konzak, C. F. (1974) A decimal code for the growth stages of cereals. *Weed Research* **14**, 415–421.

APPENDIX

Symbols used in the equations

A_{lf}	number of emerged leaves on main stem (leaves)
$A_{lf,fi}$	number of emerged leaves on main stem at floral initiation (leaves)
$A_{lf,u}$	number of unemerged leaves on main stem at floral initiation (leaves)
A_{plt}	number of plants m^{-2}
A_{prim}	number of leaf primordia on main stem (primordia)

A_{til}	number of tillers per plant (tillers plant ⁻¹)
$A_{\text{til},\text{main}}$	number of tillers on main stem per plant (tillers)
$A_{\text{til},\text{sq}}$	number of tillers m ⁻² (tillers m ⁻²)
$C_{\text{dem},\text{til}}$	assimilate demand of all the tillers (kg ha ⁻¹)
D_r	physiological development days from anthesis to maturity (PDDs)
D_s	total physiological development days from emergence to maturity (PDDs)
D_v	physiological development days from emergence to anthesis (PDDs)
d	day number (-)
d_{emerg}	emergence day of the year (1–365)
d_{fi}	day on which floral initiation occurs
d_{germ}	germination day of the year (1–365)
$f(h_{\text{php}})$	photoperiod effect on development rate (-)
$f(V)$	vernalization effect on development rate (-)
$f_r(T)$	temperature response function of development rate in post-anthesis phase (-)
$f_T(T, T_{\text{min}}, T_{\text{opt}}, T_{\text{max}})$	temperature response function with a minimum, optimum and maximum temperature used in several process (-)
$f_v(T)$	temperature response function of development rate in pre-anthesis phase (-)
$f_{\text{vn}}(T)$	temperature response function of vernalization (vernalization rate) (PVDs day ⁻¹)
h_{opp}	threshold photoperiod between optimal and non-optimal photoperiod (h)
h_{pc}	critical photoperiod at which no development occurs (h)
h_{php}	length of the natural photoperiod (astronomical day-length plus civil twilight) (h)
n	number of days (-)
PDD	physiological development day
PVD	physiological vernalization day
R_{dev}	phenological development rate (day ⁻¹)
$R_{\text{dev},\text{emerg}}$	development rate from germination to emergence (day ⁻¹)
$R_{\text{dev},r}$	actual development rate in the post-anthesis phase (day ⁻¹)
$R_{\text{dev},v}$	actual development rate in the pre-anthesis phase (day ⁻¹)
$R_{\text{dev},r,\text{max}}$	maximum development rate from anthesis to maturity (day ⁻¹)

$R_{dev,v,max}$	maximum development rate from emergence to anthesis (day^{-1})
$R_{lf,app}$	leaf appearance rate (leaves day^{-1})
$R_{lf,app,max}$	maximum leaf appearance rate (leaves day^{-1})
R_{node}	actual internode appearance rate (internodes day^{-1})
$R_{node,max}$	maximum internode appearance rate (internodes day^{-1})
$R_{prim,ini}$	actual rate of leaf primordia initiation (primordia day^{-1})
$R_{prim,max}$	maximum rate of leaf primordia initiation (primordia day^{-1})
S_{EC}	Zadoks development stages (EC stages) (–)
S_{dev}	phenological development stage (VR stage) (–)
$S_{fi,flg}$	interval of VR stage value between floral initiation and flag leaf (–)
T	temperature ($^{\circ}\text{C}$)
T_{ave}	daily average air temperature ($^{\circ}\text{C}$)
T_{base}	base temperature for emergence ($^{\circ}\text{C}$)
$T_{min,...}$	minimum temperature for a given process ($^{\circ}\text{C}$). The second subscript: v, pre-anthesis development; r, post-anthesis development; vn, vernalization; p, photosynthesis
$T_{max,...}$	maximum temperature for a given process ($^{\circ}\text{C}$). The second subscript: v, pre-anthesis development; r, post-anthesis development; vn, vernalization; p, photosynthesis
$T_{opt,...}$	optimum temperature for a given process ($^{\circ}\text{C}$). The second subscript: v, pre-anthesis development; r, post-anthesis development; vn, vernalization; lf, leaf initiation and appearance
V_n	cumulative physiological vernalization days (PVDs)
V_{nb}	base vernalization days (PVDs)
V_{nd}	vernalization requirements of a crop (saturated vernalization days) (PVDs)
$W_{total,st}$	total stem weight of the plants (kg ha^{-1})
α	parameter used in the temperature response function (–)
χ	sign variable used in the photoperiod response function (–)
$\sum T$	temperature sum needed for emergence ($^{\circ}\text{C day}$)
ω	photoperiod sensitivity of the crop (–)
ω'	photoperiod sensitivity factor (–)
ξ_{tiller}	weight of a single tiller when elongation ceases (g)